

Your LAST Name _____

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Your Collaborators

• Name: _____ SID: _____

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Policy Statement

- We encourage you to collaborate, but only in a group of up to *five* current EE 120 students.
- Be sure to write the names of your collaborators at the appropriate location on the next page. Each collaborator must submit a distinct, self-prepared solution document containing original contributions to the collaborative effort.
- To submit your work, you have several options:
 - (a) Print out this homework document and write your solutions on the printout. Then upload a *PDF file* of the scan.
 - (b) Write your solutions on this PDF file using a tablet device. Then upload a *PDF file*.
 - (c) Use your own sheets of paper, but demarcate clearly the space for each problem, part, and subpart. Then upload a *PDF file*.
- Please write neatly and legibly, because *if we can't read it, we can't evaluate it*. Remember that, even though the problem sets are self-graded, we will evaluate your work whenever we check your self grades.
- Unless we explicitly state otherwise, you may expect full credit *only if* you explain your work succinctly, but clearly and convincingly.
- If you are asked to provide a “sketch,” it refers to a well-labeled *hand-drawn* sketch—not a plot generated by a computing device.
- On occasion, a problem set contains one or more problems designated as “optional.” Your solutions to such problems will NOT be evaluated. Nevertheless, you’re responsible for learning the subject matter within their scope.
- If you have a confirmed disability that precludes you from complying fully with these instructions or with any other parameter associated with this problem set, please alert us immediately about reasonable accommodations afforded to you by the DSP Office on campus.

Overview

This problem set explores some aspects of Euler's Formula; the convolution of discrete-time signals; the system properties of linearity and time-invariance; and the impulse responses and frequency responses of discrete-time LTI (DT-LTI) systems.

HW2.1 This problem explores another consequence of Euler's Formula

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

Show that any linear combination of a set of cosine waves of frequency ω is always a cosine wave of the same frequency, *even if* each cosine wave has a distinct phase. In particular, show that

$$\sum_{k=1}^N A_k \cos(\omega t + \phi_k) = A \cos(\omega t + \phi).$$

Hint: Recognize that any complex number can be written as $Ae^{i\phi}$ for suitable A and ϕ . Further notice that $\sum_{k=1}^N A_k e^{i\phi_k}$ is a complex number.

Using the hint, equate the summation to the general complex number form.

$$\sum_{k=1}^N A_k e^{i\phi_k} = A e^{i\phi}$$

Multiply by $e^{i\omega t}$ on both sides and collect the complex exponential.

$$\sum_{k=1}^N A_k e^{i\phi_k} e^{i\omega t} = A e^{i\phi} e^{i\omega t}$$

$$\sum_{k=1}^N A_k e^{i(\omega t + \phi_k)} = A e^{i(\omega t + \phi)}$$

Now to introduce the cosines, rewrite using Euler's Formula.

$$\sum_{k=1}^N A_k (\cos(\omega t + \phi_k) + i \sin(\omega t + \phi_k)) = A (\cos(\omega t + \phi) + i \sin(\omega t + \phi))$$

$$\sum_{k=1}^N A_k \cos(\omega t + \phi_k) + i \sum_{k=1}^N A_k \sin(\omega t + \phi_k) = A \cos(\omega t + \phi) + i A \sin(\omega t + \phi)$$

Notice that the real parts and imaginary parts of both sides must be equal to each other. Focusing on just the real part, we can obtain the desired result.

$$\sum_{k=1}^N A_k \cos(\omega t + \phi_k) = A \cos(\omega t + \phi)$$

HW2.2 (Is it LTI?) Each of the following equations describes a relationship that holds between an input signal x and an output signal y of an associated discrete-time system, for all integers n . In each case, determine if the system is (i) linear and (ii) time-invariant.

Here are a few refreshers on definitions. Suppose a system maps a signal x to another signal y .

- The system is said to be *linear* if it satisfies two properties.
 1. **Scaling:** For any scalar $a \in \mathbb{C}$, ax maps to ay .
 2. **Superposition:** If x_1 maps to y_1 and x_2 maps to y_2 , then $x_1 + x_2$ maps to $y_1 + y_2$, for any signals x_1, x_2 .

One of the easiest ways to show a system is *not* linear is by showing it breaks the scaling property in the case of $a = 0$: a linear system *always* maps an all zero input signal to an all zero output signal. Conversely, if you have a hunch that a system *is* linear, it's often easier to check both properties at once by verifying that $ax_1 + bx_2$ maps to $ay_1 + by_2$, where y_1 is the system's output for x_1 and y_2 is the system's output for x_2 .

- The system is said to be *time-invariant* if $x[n - n_d]$ maps to $y[n - n_d]$ for any sample delay $n_d \in \mathbb{Z}$. That is, delaying the input to a time-invariant system will cause it to produce the same output, just delayed by that amount.

(a) $y[n] = 7x[n + 1]$.

- (i) **Linear: Yes.** Consider two input signals x_1, x_2 , whose respective outputs are $y_1[n] = 7x_1[n + 1]$ and $y_2[n] = 7x_2[n + 1]$. Now consider $x = \alpha x_1 + \beta x_2$, where $\alpha, \beta \in \mathbb{C}$. The output for this system is $y[n] = 7x[n + 1] = 7(\alpha x_1[n + 1] + \beta x_2[n + 1]) = \alpha(7x_1[n + 1]) + \beta(7x_2[n + 1]) = \alpha y_1[n] + \beta y_2[n]$, so the system is linear.
- (ii) **Time-invariant: Yes.** The system maps $x[n]$ to $y[n] = 7x[n + 1]$. Suppose we delay the input signal x by n_d samples, where $n_d \in \mathbb{Z}$, to obtain $x[n - n_d]$. Then the output is $7x[(n + 1) - n_d] = 7x[n - n_d + 1] = y[n - n_d]$. Delaying any input signal x by n_d samples delays the output by n_d samples, thus the system is time-invariant.

(b) $y[n] = x[n]x[n - 1]$

- (i) **Linear: No.** The scaling property is violated. We know $x[n]$ maps to $y[n] = x[n]x[n - 1]$. Consider $ax[n]$. This maps to $(ax[n])(ax[n - 1]) = a^2x[n]x[n - 1] = a^2y[n]$, which is, in general, not equal to $ay[n]$. For example, if $a = 3$, then $3x[n]$ maps to $9y[n]$, not $3y[n]$.

- (ii) **Time-invariant: Yes.** If we delay the input by n_d samples, the output is $x[n - n_d]x[(n - n_d) - 1] = x[n - n_d]x[n - n_d - 1] = y[n - n_d]$.
- (c) $y[n] = e^{x[n]}$
- (i) **Linear: No.** The intuition here is that the output is an exponential function of the input, which clearly isn't linear. More rigorously, if $x[n] = 0$, $y[n] = 1$, so the system violates the scaling property and can't be linear.
- (ii) **Time-invariant: Yes.** We know $y[n] = e^{x[n]}$. Now, delay the input by n_d samples to get $x[n - n_d]$. The output is now $e^{x[n - n_d]}$. But the original output delayed by n_d samples, $y[n - n_d]$, is also $e^{x[n - n_d]}$. Thus the system is time-invariant.
- (d) $y[n] = x[-n]$
- (i) **Linear: Yes.** Consider x_1, x_2 , which map to $y_1[n] = x_1[-n]$ and $y_2[n] = x_2[-n]$. Now define $x[n] = ax_1[n] + bx_2[n]$, where $a, b \in \mathbb{C}$. The corresponding output, y , is $y[n] = x[-n] = ax_1[-n] + bx_2[-n] = ay_1[n] + by_2[n]$, so the system is linear.
- (ii) **Time-invariant: No.** Here, it's easiest to consider a specific counterexample. If $x[n] = \delta[n]$, $y[n] = \delta[n]$. Suppose we shift the input to the right by one, so that we send in $x[n - 1] = \delta[n - 1]$. Then the output is $\delta[-n - 1]$, which is $y[n]$ shifted to the *left* by one!
- (e) $y[n] = v[n]x[n]$, where v is a fixed, non-constant discrete-time signal.
- (i) **Linear: Yes.** Consider x_1, x_2 , which map to $y_1[n] = v[n]x_1[n]$ and $y_2[n] = v[n]x_2[n]$. Now define $x[n] = ax_1[n] + bx_2[n]$, where $a, b \in \mathbb{C}$. The corresponding output, y , is $y[n] = v[n]x[n] = v[n](ax_1[n] + bx_2[n]) = av[n]x_1[n] + bv[n]x_2[n] = ay_1[n] + by_2[n]$, so the system is linear.
- (ii) **Time-invariant: No.** Just as with the previous system, it's easiest to give a counterexample by considering a specific case. Suppose $v[n] = \delta[n]$, and consider any input signal $x[n]$. We have $y[n] = \delta[n]x[n]$, so that $y(0) = x(0)$ and otherwise $y[n] = 0$. Now, send in $\tilde{x}[n] = x[n - 1]$. Then $\tilde{y}[n] = \delta[n]\tilde{x}[n] = \delta[n]x[n - 1]$, so $\tilde{y}[0] = x[-1]$ and otherwise $\tilde{y}[n] = 0$. Clearly, $\tilde{y}[n] \neq y[n - 1]$, so the system isn't time-invariant.

The system here is ubiquitous in digital signal processing, where v is known as a *windowing function*. It's often of interest to crop signals and focus on one part of them by masking out all irrelevant samples as zero. By setting $v[n]$ to 1 at places we want to keep the signal (almost always, these will be consecutive samples), and 0 at places we want to discard, we can apply a rectangular window, for example. In EE 123, a follow on course to EE 120, the engineering tradeoffs involved in the design of v are discussed in more detail.

HW2.3 The impulse response of a discrete-time LTI system F



is the *discrete-time doublet*: $f[n] = \frac{1}{2}[\delta[n] - \delta[n - 1]]$.

- (a) For each of the inputs below, determine the corresponding output of the system. Be sure to provide a well-labeled sketch of all the signals involved.

First, it is useful to solve for $y[n]$ in terms of $x[n]$ using the convolution formula:

$$\begin{aligned}
 y[n] &= (x * f)[n] \\
 &= \sum_{k=-\infty}^{\infty} f[k]x[n - k] \\
 &= \frac{1}{2} \sum_{k=-\infty}^{\infty} (\delta[k]x[n - k] - \delta[k - 1]x[n - k])
 \end{aligned}$$

Because $\delta[k]$ is 1 for $k = 0$ and zero elsewhere, $\delta[k]x[n - k]$ is equivalent to $\delta[k]x[n - 0]$. Likewise, $\delta[k - 1]x[n - k] = \delta[k - 1]x[n - 1]$. This is known as the **sampling property** of the Kronecker delta function. So,

$$\begin{aligned}
 y[n] &= \frac{1}{2} \sum_{k=-\infty}^{\infty} (\delta[k]x[n] - \delta[k - 1]x[n - 1]) \\
 &= \frac{1}{2}[x[n] - x[n - 1]].
 \end{aligned}$$

Note that this is equivalent to replacing $f[n]$ with $y[n]$ and $\delta[n]$ with $x[n]$ in the expression $f[n] = \delta[n] - \delta[n - 1]$.

- (i) The unit-step function: $x[n] = u[n]$ where $u[n] = \begin{cases} 1 & \text{if } n \geq 0 \\ 0 & \text{if } n < 0 \end{cases}$.

$$y[n] = \frac{1}{2}[x[n] - x[n - 1]] = \frac{1}{2}[u[n] - u[n - 1]] = \frac{1}{2}\delta[n].$$

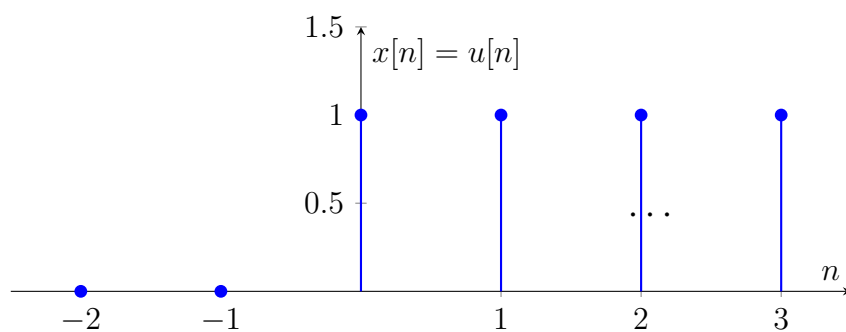


Figure 1: Plot of $x[n]$ for 1.2(a)(i).

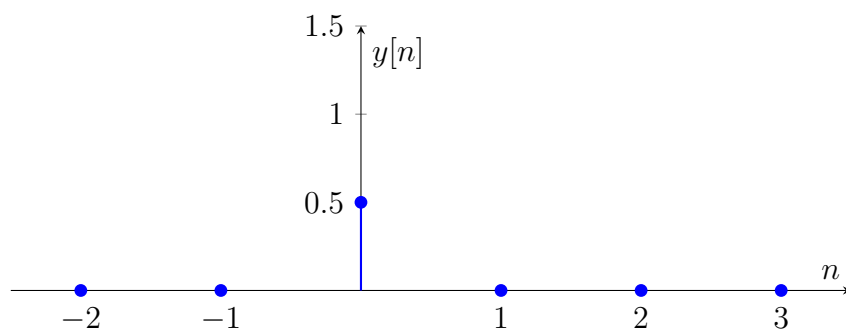


Figure 2: Plot of $y[n]$ for 1.2(a)(i).

(ii) A four-point discrete-time box function:

$$x[n] = \delta[n] + \delta[n - 1] + \delta[n - 2] + \delta[n - 3].$$

$$\begin{aligned} y[n] &= \frac{1}{2}[x[n] - x[n - 1]] \\ &= \frac{1}{2}[\delta[n] + \delta[n - 1] + \delta[n - 2] + \delta[n - 3] \\ &\quad - \delta[n - 1] - \delta[n - 2] - \delta[n - 3] - \delta[n - 4]] \\ &= \frac{1}{2}[\delta[n] - \delta[n - 4]] \end{aligned}$$

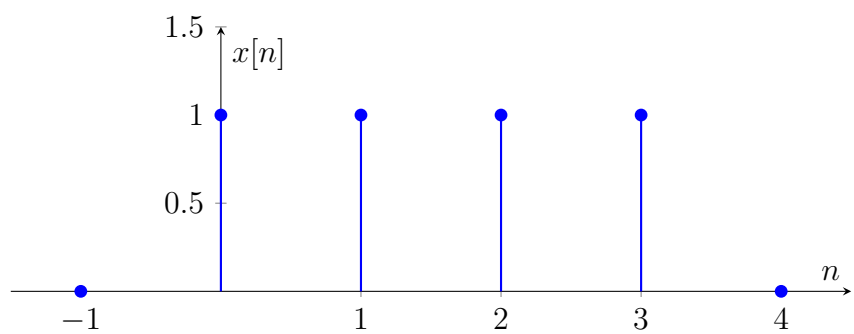


Figure 3: Plot of $x[n]$ for 1.2(a)(ii).

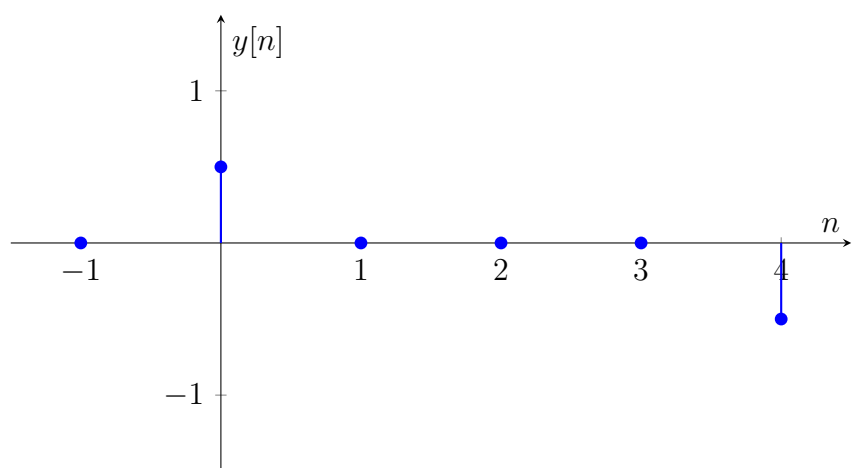


Figure 4: Plot of $y[n]$ for 1.2(a)(ii).

(iii) A constant function: $x[n] = 1$.

$$\begin{aligned} y[n] &= \frac{1}{2}(x[n] - x[n-1]) \\ &= \frac{1}{2}(1 - 1) = 0 \end{aligned}$$

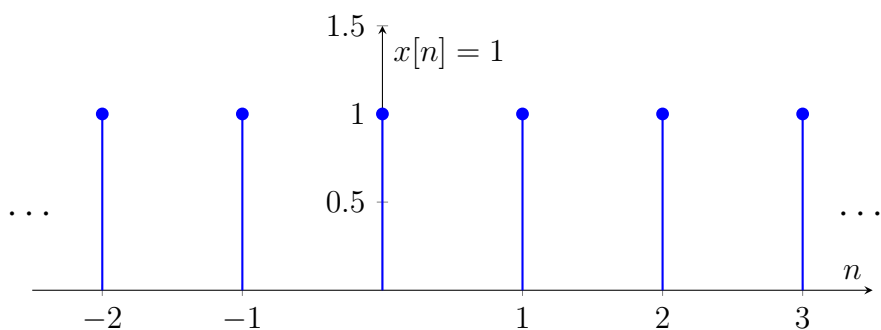


Figure 5: Plot of $x[n]$ for 1.2(a)(iii).

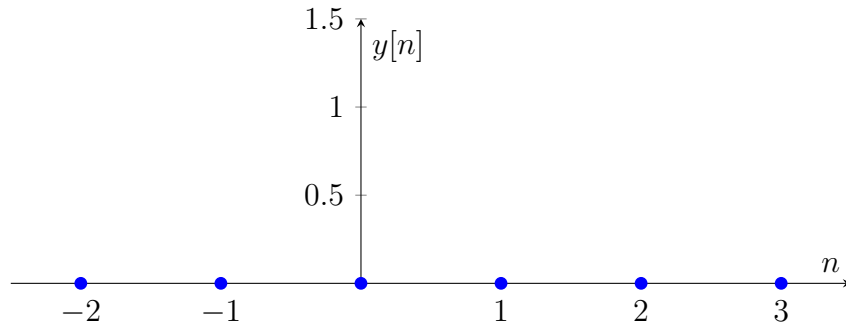


Figure 6: Plot of $y[n]$ for 1.2(a)(iii).

(iv) The sign-alternating signal $x[n] = 2(-1)^n$.

$$\begin{aligned}
 y[n] &= \frac{1}{2}(x[n] - x[n-1]) \\
 &= (-1)^n - (-1)^{n-1} \\
 &= (-1)^n - \frac{(-1)^n}{-1} = 2(-1)^n
 \end{aligned}$$

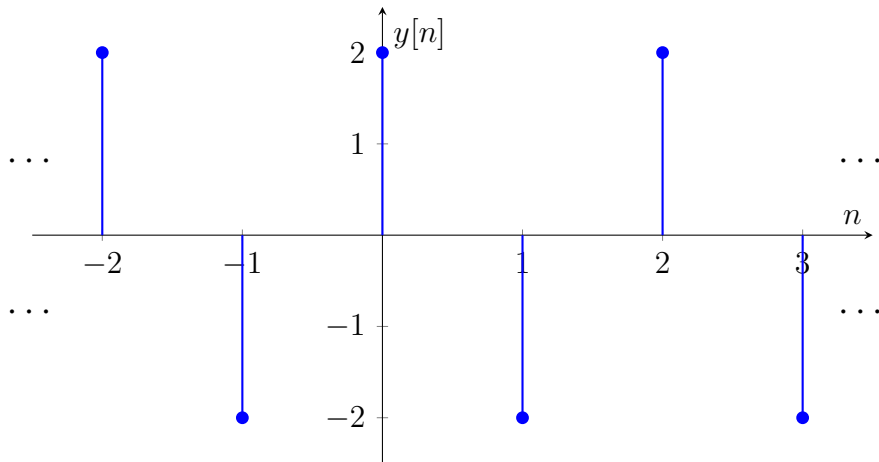
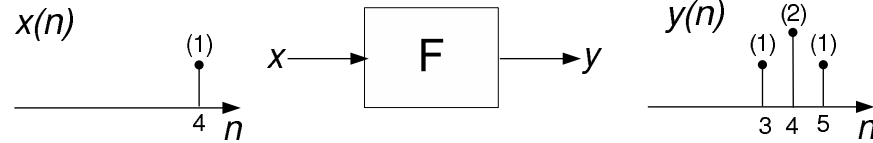


Figure 7: Plot of $y[n]$ for 1.2(a)(iv).

(b) Based on the results of part (a), explain why the system F may aptly be called a one-dimensional *edge detector*. Does the system's response provide a clue as to which direction (up or down) each edge in the input signal is?

As seen in the figures, we note that the output is nonzero whenever the signal changes. When the input increases with time, the output is positive and vice versa. This gives an indication of the direction of the edge in the input signal.

HW2.4 Consider a discrete-time system $F : [\mathbb{Z} \rightarrow \mathbb{R}] \rightarrow [\mathbb{Z} \rightarrow \mathbb{R}]$. It is known that F is linear and time invariant. The following input-output signal pair (x, y) is a behavior of the system. The signals x and y are zero outside the regions shown.



Provide succinct, but clear and convincing responses to the questions below.

- (a) Can the impulse response f be determined from the information given in this problem? If so, provide a well-labeled plot of the sample values $f[n]$. If not, explain why it is not possible to determine f based on what is known about the system F .

Yes, it can. We know the system is time-invariant, so if we send in $x[n + 4]$ as the input, we know that the corresponding output must be $y[n + 4]$. Since $x[n + 4] = \delta[n]$, we know that the impulse response must be $y[n + 4]$. (The impulse response of an LTI system is defined as the output signal that we get when the input signal is the Kronecker delta.) See Figure 8 for a plot of the impulse response $f[n]$.

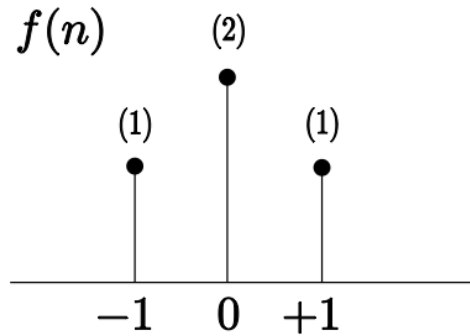


Figure 8: Plot of $f[n]$ for 1.3(a).

- (b) Determine, and provide well-labeled, hand-drawn sketches of, the response of the system to each of the following input signals:

(i) $x[n] = 1$ for all $n \in \mathbb{Z}$. First, let's solve for $y[n]$ in terms of $x[n]$.

$$\begin{aligned}
 y[n] &= (x * f)[n] \\
 &= \sum_{k=-\infty}^{\infty} f[k]x[n-k] \\
 &= \sum_{k=-\infty}^{\infty} (\delta[k+1] + 2\delta[k] + \delta[k-1])x[n-k] \\
 &= x[n+1] + 2x[n] + x[n-1],
 \end{aligned}$$

where the last step applies the **sifting property** of the Kronecker delta. Generically speaking,

$$\sum_{k=A}^B \delta[k-N]h[k] = h[N],$$

where $A < N < B$. Applying the formula for $y[n]$,

$$\begin{aligned}
 y[n] &= x[n+1] + 2x[n] + x[n-1] \\
 &= 4
 \end{aligned}$$

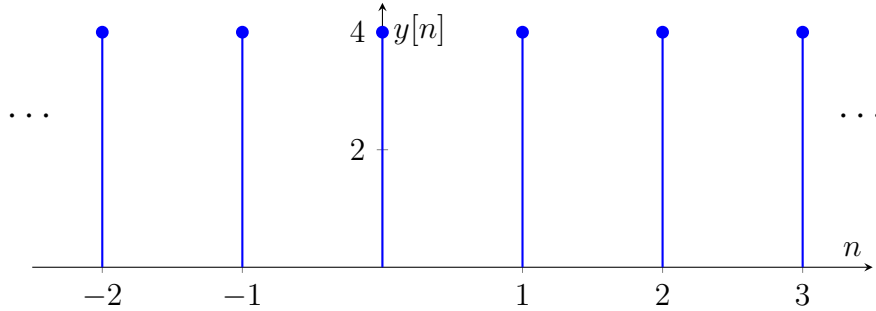


Figure 9: Plot of $y[n]$ for 1.3(b) i.

(ii) $x[n] = \cos(\pi n)$ for all $n \in \mathbb{Z}$.

$$\begin{aligned}
 x[n] &= \cos(\pi n) = \frac{e^{i\pi n} + e^{-i\pi n}}{2} \\
 &= e^{i\pi n} = (-1)^n \\
 y[n] &= x[n+1] + 2x[n] + x[n-1] \\
 &= (-1)^{n+1} + 2(-1)^n + (-1)^{n-1} \\
 &= (-1)(-1)^n + 2(-1)^n + \frac{(-1)^n}{-1} \\
 &= (-1)^n(-1 + 2 - 1) \\
 &= 0
 \end{aligned}$$

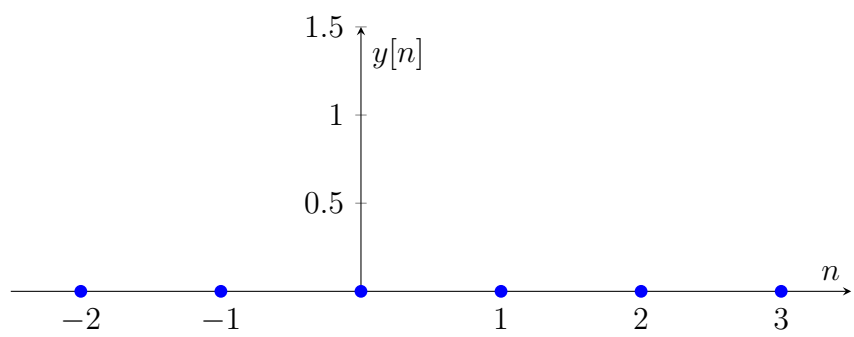


Figure 10: Plot of $y[n]$ for 1.3(b) ii.

HW2.5 (Discrete convolution practice) Consider the signal

$$x[n] = \alpha^n u[n].$$

- (a) Sketch the signal $g[n] = x[n] - \alpha x[n - 1]$. Write $g[n]$ as a convolution related to $x[n]$.

$$g[n] = x[n] - \alpha x[n - 1] = \alpha^n u[n] - \alpha^n u[n - 1] = \delta[n].$$

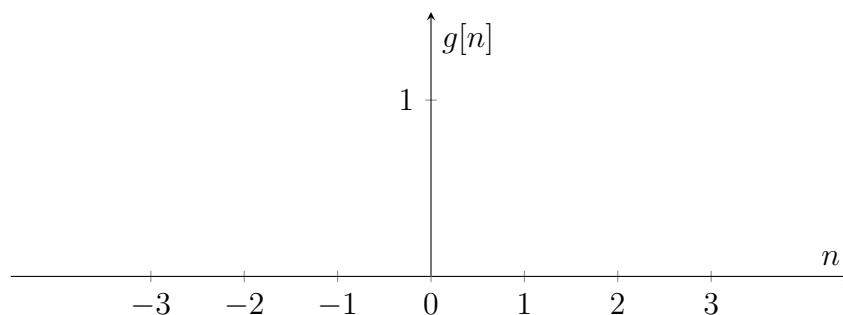


Figure 11

Using the fact of $x[n] * \delta[n] = x[n]$ and $x[n] * \delta[n - 1] = x[n - 1]$, we can write $g[n] = x[n] * (\delta[n] - \alpha \delta[n - 1]) = \delta[n]$.

- (b) Use the result of part (a) along with the properties of convolution to determine $h[n]$ such that

$$x[n] * h[n] = \left(\frac{1}{2}\right)^n (u[n + 2] - u[n - 2]).$$

Hint: Use the fact that $x[n] * \delta[n] = x[n]$.

From (a), we know that $g[n] = x[n] * (\delta[n] - \alpha \delta[n - 1]) = \delta[n]$. We also know $x[n] * h[n] = \left(\frac{1}{2}\right)^n (u[n + 2] - u[n - 2]) = y[n]$ and $\delta[n] * y[n] = y[n]$. So

$$x[n] * (\delta[n] - \alpha \delta[n - 1]) * y[n] = y[n]$$

$$\begin{aligned} h[n] &= (\delta[n] - \alpha \delta[n - 1]) * y[n] \\ &= (\delta[n] - \alpha \delta[n - 1]) * \left(\left(\frac{1}{2}\right)^n (u[n + 2] - u[n - 2])\right) \\ &= \left(\frac{1}{2}\right)^n (u[n + 2] - u[n - 2]) - \frac{\alpha}{2^{n-1}} (u[n + 1] - u[n - 3]) \\ &= 4\delta[n + 2] + (2 - 4\alpha)\delta[n + 1] + (1 - 2\alpha)\delta[n] \\ &\quad + \left(\frac{1}{2} - \alpha\right)\delta[n - 1] - \frac{1}{2}\alpha\delta[n - 2] \end{aligned}$$

HW2.6 (Polynomial Multiplication and Discrete-Time Convolution) The impulse response of a real¹, discrete-time FIR filter A is described by

$$a[n] = a_0\delta[n] + a_1\delta[n - 1] + \cdots + a_N\delta[n - N], \quad \text{where } N \in \{1, 2, 3, \dots\}.$$

- (a) Show that the frequency response $A(\omega)$ of the filter is a polynomial, in terms of $e^{-i\omega}$, whose coefficients have a simple relationship with the impulse response values $a_0, \dots, a_N$.

Substituting $a(n)$ into the frequency response formula we see that:

$$A(w) = \sum_{n=-\infty}^{\infty} a[n]e^{-iwn}$$

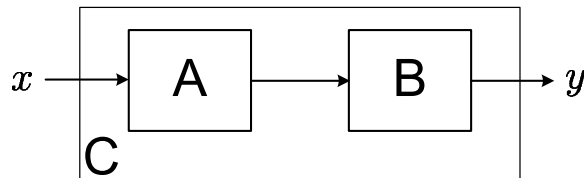
As $a(n) = 0$ outside the range of $n = 0$ to $n = N$,

$$\begin{aligned} A(w) &= a_0e^{-i\omega \cdot 0} + a_1e^{-i\omega \cdot 1} + \dots + a_Ne^{-i\omega \cdot N} \\ &= a_0 + a_1e^{-i\omega} + \dots + a_Ne^{-i\omega N} \end{aligned}$$

- (b) Suppose the filter A is placed in a cascade (series) interconnection with another discrete-time FIR filter B whose impulse response is described by

$$b[n] = b_0\delta[n] + b_1\delta[n - 1] + \cdots + b_M\delta[n - M], \quad \text{where } M \in \{1, 2, 3, \dots\}.$$

The cascade structure, which we call C, is shown in the figure below.



Let c denote the impulse response of the cascade interconnection.

- (i) Find an expression for $c[n]$ in terms of $a_0, a_1, \dots, a_N$ and $b_0, b_1, \dots, b_M$.

The impulse response c of the cascade interconnection is given by

$$c[n] = (a * b)[n] = \sum_{k=-\infty}^{\infty} a[k]b[n - k]$$

Substituting $n = 0$:

$$c_0 = a_0b_0$$

¹A real filter is one that produces a real-valued output signal for every real-valued input signal. An LTI filter is real if its impulse-response is real-valued.

n = 1:

$$c_1 = a_0b_1 + a_1b_0$$

n = 2:

$$c_2 = a_0b_2 + a_1b_1 + a_2b_0$$

And so on...

Thus, we can write:

$$c_n = \sum_{k=\max(0, n-M)}^{\min(n, N)} a_k b_{n-k}$$

for n = 0, 1, ..., N + M

where we found the bounds of summation from the two conditions that ensure we are indexing $a(n)$ and $b(n)$ without exceeding the bounds of their signal.

$$0 \leq k \leq N \implies k \leq N \quad \text{and} \quad k \geq 0$$

$$0 \leq n - k \leq M \implies k \leq n \quad \text{and} \quad k \geq n - M$$

and so

$$k \leq \min(n, N) \quad \text{and} \quad k \geq \max(0, n - M)$$

Lastly,

$$c[n] = c_0\delta[n] + c_1\delta[n-1] + \dots + c_{N+M}\delta[n-(N+M)]$$

(ii) Consider two polynomials $A(z)$ and $B(z)$ described as follows:

$$A(z) = a_0 + a_1z + \cdots + a_Nz^N \quad B(z) = b_0 + b_1z + \cdots + b_Mz^M,$$

where M and N are positive integers. Let $C(z) = A(z) B(z)$, where

$$C(z) = c_0 + c_1z + \cdots + c_{N+M}z^{N+M}.$$

Show that multiplying polynomials is tantamount to convolving their coefficients; in particular, explain how $c_n = (a * b)_n = \sum_m a_m b_{n-m}$, where $n = 0, 1, \dots, N + M$.

$$C(z) = (a_0 + a_1z + \cdots + a_Nz^N)(b_0 + b_1z + \cdots + b_Mz^M)$$

$$C(z) = a_0b_0 + (a_0b_1 + a_1b_0)z + (a_0b_2 + a_1b_1 + a_2b_0)z^2 + \dots$$

As you can see, each coefficient of z^n is the sum of all products of a_i and b_j where $i + j = n$:

$$c_n = \sum_{m=\max(0, n-M)}^{\min(n, N)} a_m b_{n-m}$$

for $n = 0, 1, 2, \dots, N + M$

Just like the definition of convolution!